

MSE 360: Kinetic Processes in Materials

Homework #8

Instructions: Answer the following questions on a separate piece of paper. Be sure that your name and date are on the first page. Upload your work as a PDF to Moodle for credit. *All work must be done individually.*

1. Based on your knowledge of the approximate order of magnitude for gas vs. liquid vs. solid-state diffusivity, identify the most likely (only one) value for diffusivity for each of the following situations:
 - a. The self-diffusion coefficient of water molecules in water at room temperature is:
 - i. $10^{-16} \text{ cm}^2/\text{s}$
 - ii. $10^{-8} \text{ cm}^2/\text{s}$
 - iii. $10^{-4} \text{ cm}^2/\text{s}$
 - iv. $0.1 \text{ cm}^2/\text{s}$
 - v. $100 \text{ cm}^2/\text{s}$
 - b. The diffusion coefficient of water molecules in air at 28°C is about:
 - i. $10^{-16} \text{ cm}^2/\text{s}$
 - ii. $10^{-8} \text{ cm}^2/\text{s}$
 - iii. $10^{-4} \text{ cm}^2/\text{s}$
 - iv. $0.1 \text{ cm}^2/\text{s}$
 - v. $100 \text{ cm}^2/\text{s}$
 - c. The diffusion coefficient of oxygen vacancies in solid ZrO_2 at 300 K is about:
 - i. $10^{-16} \text{ cm}^2/\text{s}$
 - ii. $10^{-8} \text{ cm}^2/\text{s}$
 - iii. $10^{-4} \text{ cm}^2/\text{s}$
 - iv. $0.1 \text{ cm}^2/\text{s}$
 - v. $100 \text{ cm}^2/\text{s}$
 - d. The diffusion coefficient of oxygen vacancies in solid ZrO_2 at 1000 K is about:
 - i. $10^{-16} \text{ cm}^2/\text{s}$
 - ii. $10^{-8} \text{ cm}^2/\text{s}$
 - iii. $10^{-4} \text{ cm}^2/\text{s}$
 - iv. $0.1 \text{ cm}^2/\text{s}$
 - v. $100 \text{ cm}^2/\text{s}$
 - e. The diffusion coefficient of Cu atoms in liquid copper at its melting point is about:
 - i. $10^{-16} \text{ cm}^2/\text{s}$
 - ii. $10^{-8} \text{ cm}^2/\text{s}$
 - iii. $10^{-4} \text{ cm}^2/\text{s}$
 - iv. $0.1 \text{ cm}^2/\text{s}$
 - v. $100 \text{ cm}^2/\text{s}$

2. The experimentally measured diffusion coefficients as a function of temperature for hydrogen diffusion in SiO₂ are given in Table 4.5. Using a graphing software (e.g. Excel, Origin), answer the following questions.

TABLE 4.5 Hydrogen Diffusion in SiO₂

T (K)	D (cm ² /s)	T (K)	D (cm ² /s)	T (K)	D (cm ² /s)
373	8.10×10^{-10}	873	1.50×10^{-6}	1373	1.30×10^{-5}
473	9.99×10^{-9}	973	4.21×10^{-6}	1473	1.68×10^{-5}
573	3.22×10^{-8}	1073	4.54×10^{-6}	1573	3.32×10^{-5}
673	2.58×10^{-7}	1173	6.85×10^{-6}	1673	2.56×10^{-5}
773	3.20×10^{-7}	1273	7.19×10^{-6}	1773	2.85×10^{-5}

- Provide a plot of $\log D$ versus $1/T$ (K⁻¹).
- From these data, calculate ΔG_{act} (kJ/mol).
- From these data, calculate D_0 (cm²/s).
- Calculate the value of D (cm²/s) at 27°C.